Crystal Bases: Talk 10

Schensted Insertion, RSK and Crystals

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Last Time...

- Plactic Monoid
- Knuth Equivalence ⇒ Plactic Equivalence

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This Time…

- A (Seemingly) unrelated combinatorial strategy
- Correspondence of pairs of Semi-standard / Standard Young Tableaux (SSYT / SYT)
- Hopefully… some connection back to Crystals

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Schensted

- Throughout this talk, fix n, k positive natural number
- The *Schensted Insertion* takes a SSYT T and $i \in [n]$, creates a table denoted by $T \leftarrow i$

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The Algorithm

def insertion(T, `i`):
    set `row_count` = 1
    while true:
        if `i` ≥ the rightmost element in row `row_count`: (true if no element)
            insert `i` to the rightmost of the table
            break
        else
            find `j` such that `i` < `j` in row `row_count`
            swap `i=j`
            increase `row_count` by 1 (if no such row, create new row)
    return new table

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A demo

- $i = 4$

1	1	1	2	2	4	4
2	2	3	3	4		
3	3					
4						

6

A demo

- $i = 4$

4

1	1	1	2	2	4	4
2	2	3	3	4		
3	3					
4						

7

A demo

4

- $i = 4$

1	1	1	2	2	4	4
2	2	3	3	4		
3	3					
4						

8

A demo

- $i = 4$

1	1	1	2	2	4	4	4
2	2	3	3	4			
3	3						
4							

9

A demo

- $i = 1$

1	1	1	2	2	4	4
2	2	3	3	4		
3	3					
4						

10

A demo

- $i = 1$

1

1	1	1	2	2	4	4
2	2	3	3	4		
3	3					
4						

11

A demo

- $i = 1$

1

1	1	1	2	2	4	4
2	2	3	3	4		
3	3					
4						

12

A demo

- $i = 1$

2

1	1	1	1	2	4	4
2	2	3	3	4		
3	3					
4						

13

A demo

2 is bumped by 1

- $i = 1$

2

1	1	1	1	2	4	4
2	2	3	3	4		
3	3					
4						

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A demo

- $i = 1$

2

1	1	1	1	2	4	4
2	2	3	3	4		
3	3					
4						

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A demo

- $i = 1$

1	1	1	1	2	4	4
2	2	3	3	4		
3	3					
4						

2

16

A demo

- $i = 1$

1	1	1	1	2	4	4
2	2	2	3	4		
3	3					
4						

3

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A demo

- $i = 1$

3

1	1	1	1	2	4	4
2	2	2	3	4		
3	3					
4						

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A demo

- $i = 1$

1	1	1	1	2	4	4
2	2	2	3	4		
3	3	3				
4						

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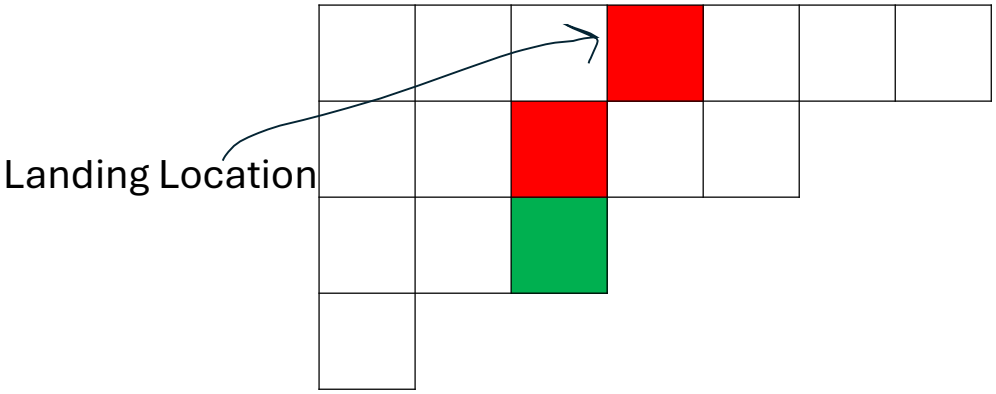
A demo

- $i = 1$

1	1	1	1	2	4	4
2	2	2	3	4		
3	3	3				
4						

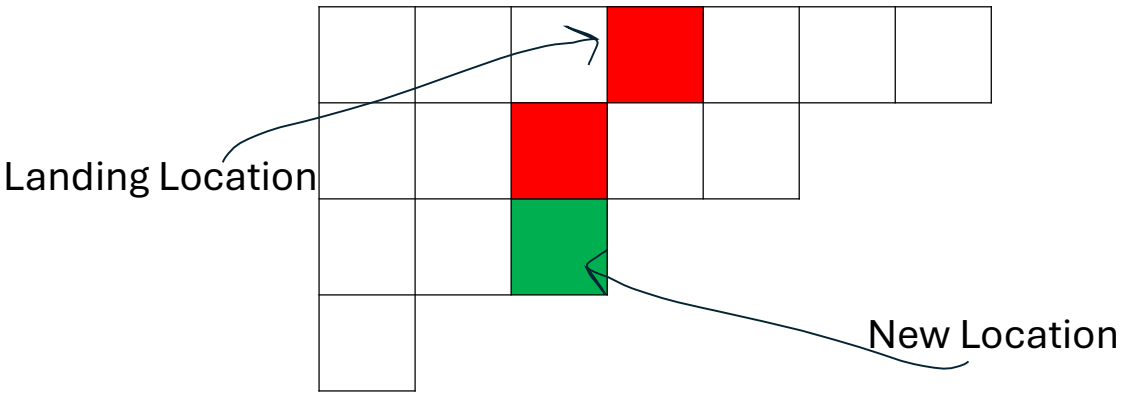
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Notation



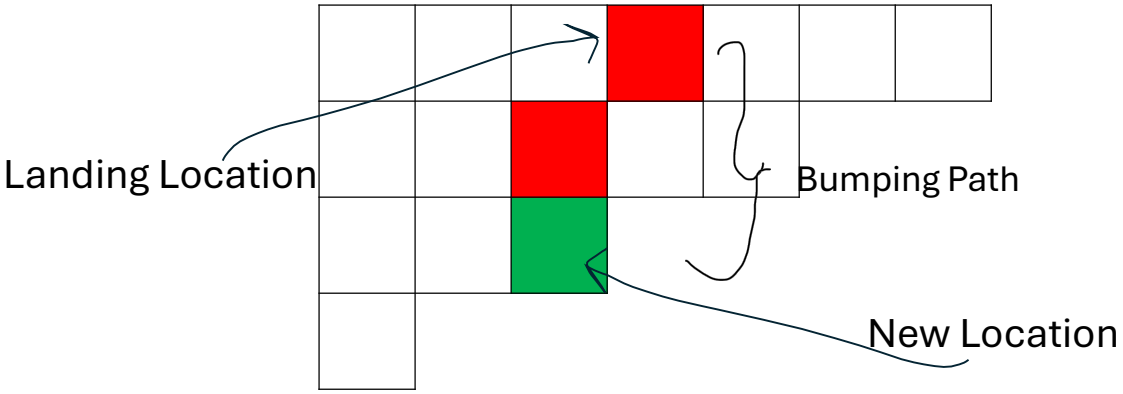
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Notation



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Notation



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Theorem

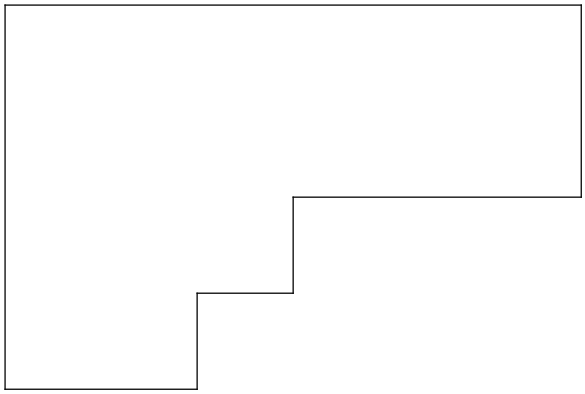
- If T is a SSYT, so is $T \leftarrow i$.

- In particular, the empty set is a SSYT, so any insertion $\emptyset \leftarrow w_1 \leftarrow w_2 \leftarrow \cdots \leftarrow w_k$ will be a SSYT as well.

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Proof

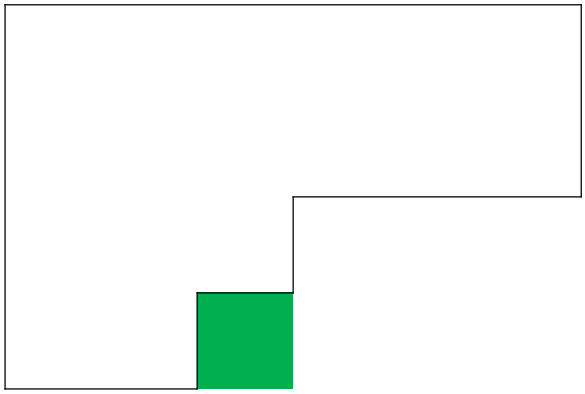
Let's Think about a general SSYT T...



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Proof

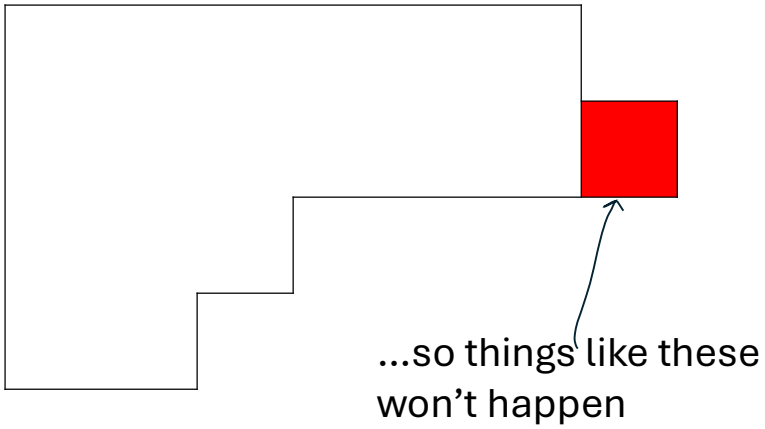
Convince yourself that any inserting keeps weakly increasing rows.



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Proof

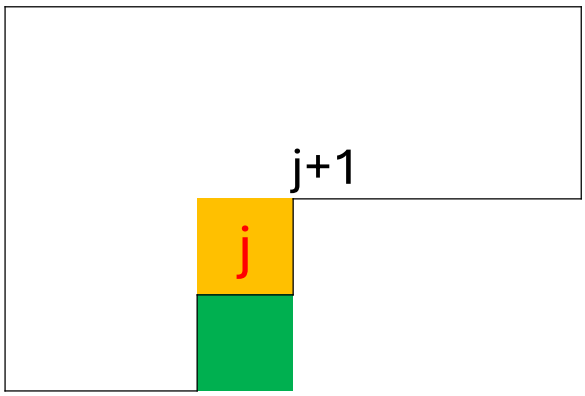
The result of insertion is also of semi-standard shape…



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Proof

Maybe it is harder that any inserting keeps *strictly* increasing columns.



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The Robinson-Schensted-Knuth (RSK) Map

- There is a bijection between

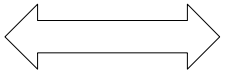


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The Robinson-Schensted-Knuth (RSK) Map

- There is a bijection between

Words of length k in alphabet $[n]$

$(w_1, w_2, \ldots w_k)$ 

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The Robinson-Schensted-Knuth (RSK) Map

- There is a bijection between

Words of length k in alphabet $[n]$

A pair (P, Q) of SSYT with alphabet $[n]$, SYT with alphabet $[k]$, with the same shape

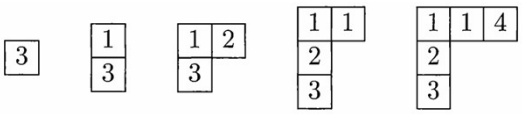
$(w_1, w_2, \ldots w_k)$ ⟺ (P, Q)

RSK Map

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Brief Description For One Direction

- Consider the word (3, 1, 2, 1, 4)



The last tableau here is P. The recording tableau Q shows where new boxes are added in the succession of insertions.

$$Q = \begin{array}{|c|c|c|c|}\hline 1 & 3 & 5 \\ \hline 2 \\ \cline{1-1} 4 \\ \cline{1-1} \end{array}\ .$$

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A Generalization

- There is a bijection between

Words of length k in alphabet $[n]$

A pair (P, Q) of SSYT with alphabet $[n]$, SYT with alphabet $[k]$, with the same shape

$(w_1, w_2, \ldots w_k)$ ⟺ (P, Q)

RSK Map

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A Generalization

- There is a bijection between

A pair (P, Q) of SSYT with alphabet $[n]$, SSYT with alphabet $[r]$, with the same shape

⟺ (P, Q)

RSK Map

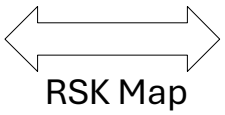
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A Generalization

- There is a bijection between

An matrices with entries in $\mathbb{N}$ with size $r \times n$

$$X = \begin{pmatrix} x_{11} & x_{12} & x_{13} \\ x_{21} & x_{22} & x_{23} \end{pmatrix}$$



RSK Map

A pair (P, Q) of SSYT with alphabet $[n]$, SSYT with alphabet $[r]$, with the same shape

$$(P, Q)$$

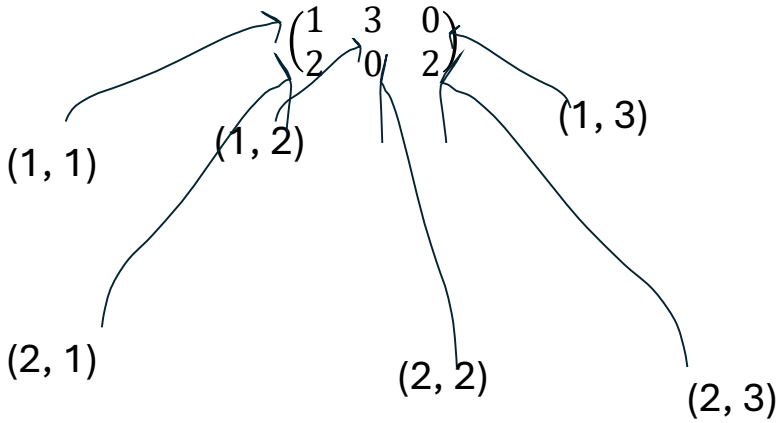
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An Example of $X \rightarrow (P, Q)$

$$\begin{pmatrix} 1 & 3 & 0 \\ 2 & 0 & 2 \end{pmatrix}$$

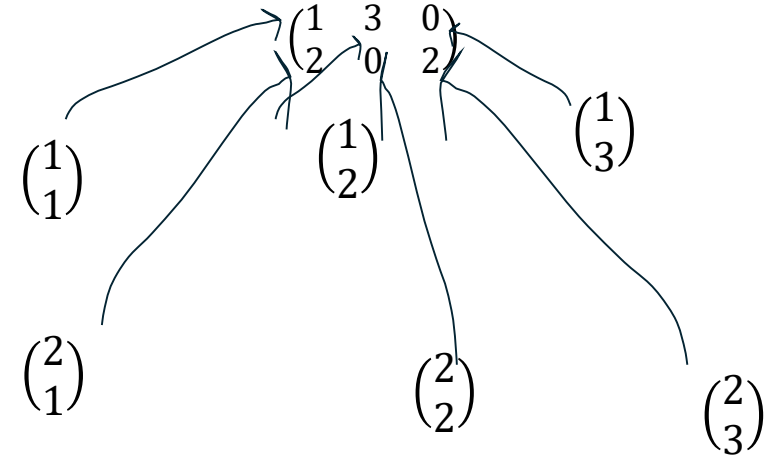
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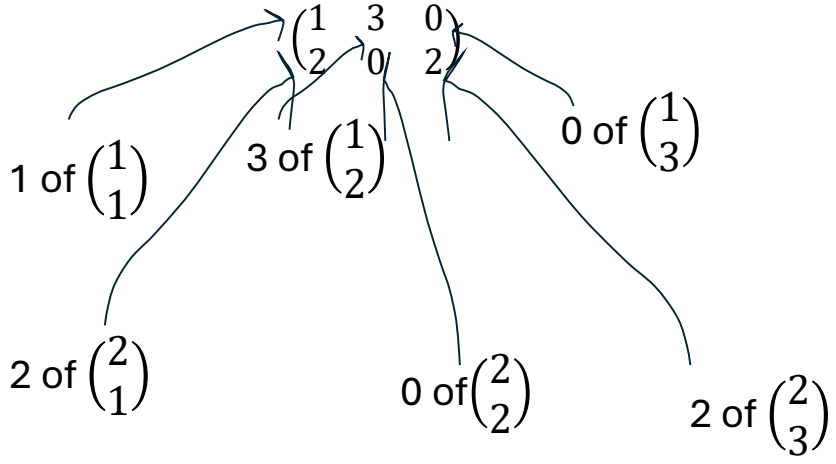
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An Example of $X \rightarrow (P, Q)$



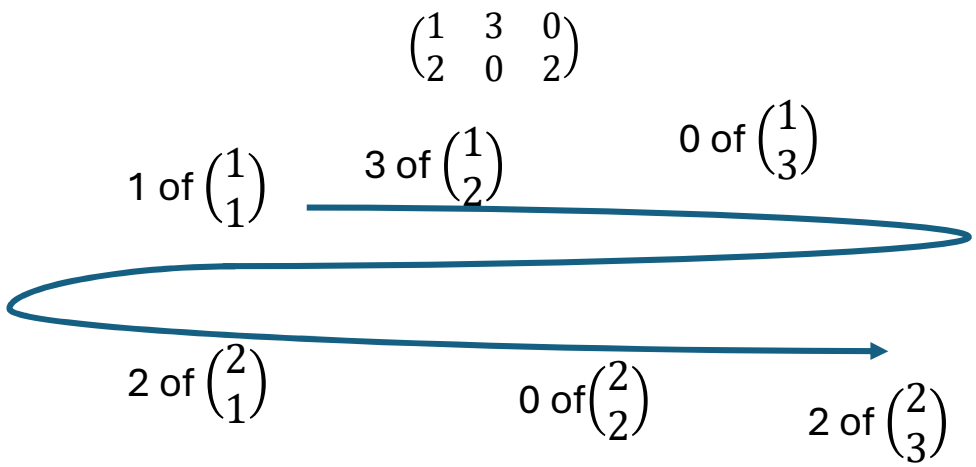
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An Example of $X \rightarrow (P, Q)$



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An Example of $X \rightarrow (P, Q)$



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An Example of $X \rightarrow (P, Q)$

$$\begin{pmatrix} 1 & 3 & 0 \\ 2 & 0 & 2 \end{pmatrix}$$

1 of $\binom{1}{1}$ 3 of $\binom{1}{2}$ 0 of $\binom{1}{3}$

Lexicographic Order

2 of $\binom{2}{1}$ 0 of $\binom{2}{2}$ 2 of $\binom{2}{3}$

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An Example of $X \rightarrow (P, Q)$

$$\begin{pmatrix} 1 & 3 & 0 \\ 2 & 0 & 2 \end{pmatrix}$$

1 of $\binom{1}{1}$ 3 of $\binom{1}{2}$ 0 of $\binom{1}{3}$

Lexicographic Order

Use these info to get another matrix A.

2 of $\binom{2}{1}$ 0 of $\binom{2}{2}$ 2 of $\binom{2}{3}$

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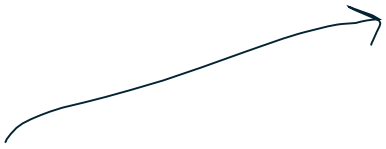
A is called a *two-rowed array*

$$A = \begin{pmatrix} 1 & 1 & 1 & 1 & 2 & 2 & 2 & 2 \\ 1 & 2 & 2 & 2 & 1 & 1 & 3 & 3 \end{pmatrix}$$

Note: You can definitely recover X from here.

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$$A = \begin{pmatrix} 1 & 1 & 1 & 1 & 2 & 2 & 2 & 2 \\ 1 & 2 & 2 & 2 & 1 & 1 & 3 & 3 \end{pmatrix}$$



Bottom Row: Get P by usual insertion. i.e. $\emptyset \leftarrow 1 \leftarrow 2 \leftarrow 2 \leftarrow 2 \ldots 3 \leftarrow 3$

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$$A = \begin{pmatrix} 1 & 1 & 1 & 1 & 2 & 2 & 2 & 2 \\ 1 & 2 & 2 & 2 & 1 & 1 & 3 & 3 \end{pmatrix}$$

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Top Row: Get Q by filling in the top row according to P

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Computing P, Q for this A

Constructing P	1	1 2	1 2 2	1 2 2 2	1 1 2 2 2	1 1 1 2 2 2	1 1 1 2 3 2 2	1 1 1 2 3 3 2 2	
bottom row of A (insertion)	1	2	2	2	1	1	3	3	
Constructing Q	1	1 1	1 1 1	1 1 1 1	1 1 1 1 2	1 1 1 1 2 2	1 1 1 1 2 2 2	1 1 1 1 2 2 2 2	
Top row of A	1	1	1	1	2	2	2	2	

Remarks:
- Constructing of P, we use red to highlight the new location, not necessarily the number inserted
- Constructing of Q, we put the number (in green) on the top row of A according to the new position

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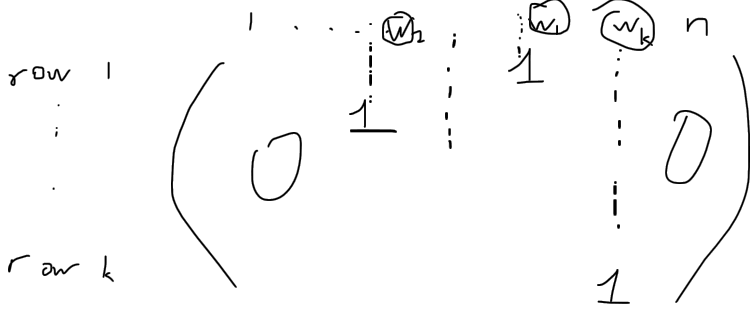
Why is it a generalization?

- Because the $k \times n$ matrix can be taken to be: given $(w_1, w_2, \ldots w_k)$:

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Why is it a generalization?

- Because the $k \times n$ matrix can be taken to be: given $(w_1, w_2, \ldots w_k)$:



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$$A = \begin{pmatrix} 1 & 2 & 3 & 4 & \cdots & \cdots & k \\ w_1 & w_2 & \cdots & \cdots & \cdots & \cdots & w_k \end{pmatrix}$$

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More notations!

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 & \cdots & \cdots & k \\ w_1 & w_2 & \cdots & \cdots & \cdots & \cdots & w_k \end{pmatrix}$$

Bottom Row: The SSYT P is usually called the **insertion tableau**

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More notations!

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 & \cdots & \cdots & k \\ w_1 & w_2 & \cdots & \cdots & \cdots & \cdots & w_k \end{pmatrix}$$

Bottom Row: The SSYT P is usually called the **insertion tableau**

Top Row: The SYT/SSYT Q is usually called the **recording tableau**

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Proof of the Generalization

- There is a bijection between

An matrices with entries in $\mathbb{N}$ with size $r \times n$

$$X = \begin{pmatrix} x_{11} & x_{12} & x_{13} \\ x_{21} & x_{22} & x_{23} \end{pmatrix}$$

RSK Map

Target

A pair (P, Q) of SSYT with alphabet $[n]$, SSYT with alphabet $[r]$, with the same shape

(P, Q)

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(Partial) Construction of Inverse

- From (P, Q), do a descent strategy:

get $\left[(P', Q'), \binom{i}{j}\right]$, which will be the [All but last, last] column of the resulting two-rowed array

descent…

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(Partial) Construction of Inverse

- (Sketch) From (P, Q), we know that $P = P' \leftarrow j$, so if we get P', that recovers Q', i

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(Partial) Construction of Inverse

- (Sketch) From (P, Q), we know that $P = P' \leftarrow j$, so if we get P', that recovers Q', i
  - Suffice to find the bumping path of the insertion.

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(Partial) Construction of Inverse

- (Sketch) From (P, Q), we know that $P = P' \leftarrow j$, so if we get P', that recovers Q', i
  - Suffice to find the bumping path of the insertion.
    - New Location: Can be determined by Q alone.

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(Partial) Construction of Inverse

- (Sketch) From (P, Q), we know that $P = P' \leftarrow j$, so if we get P', that recovers Q', i
  - Suffice to find the bumping path of the insertion.
    - New Location: Can be determined by Q alone.
    - Knowing P and the new location, we can determine the bumping path.

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How about Crystals?



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